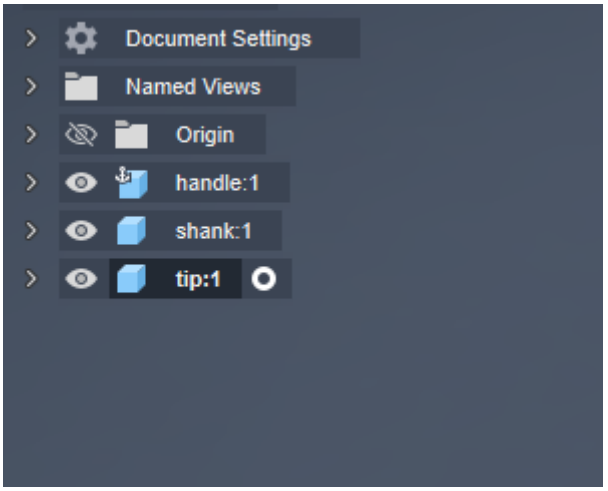
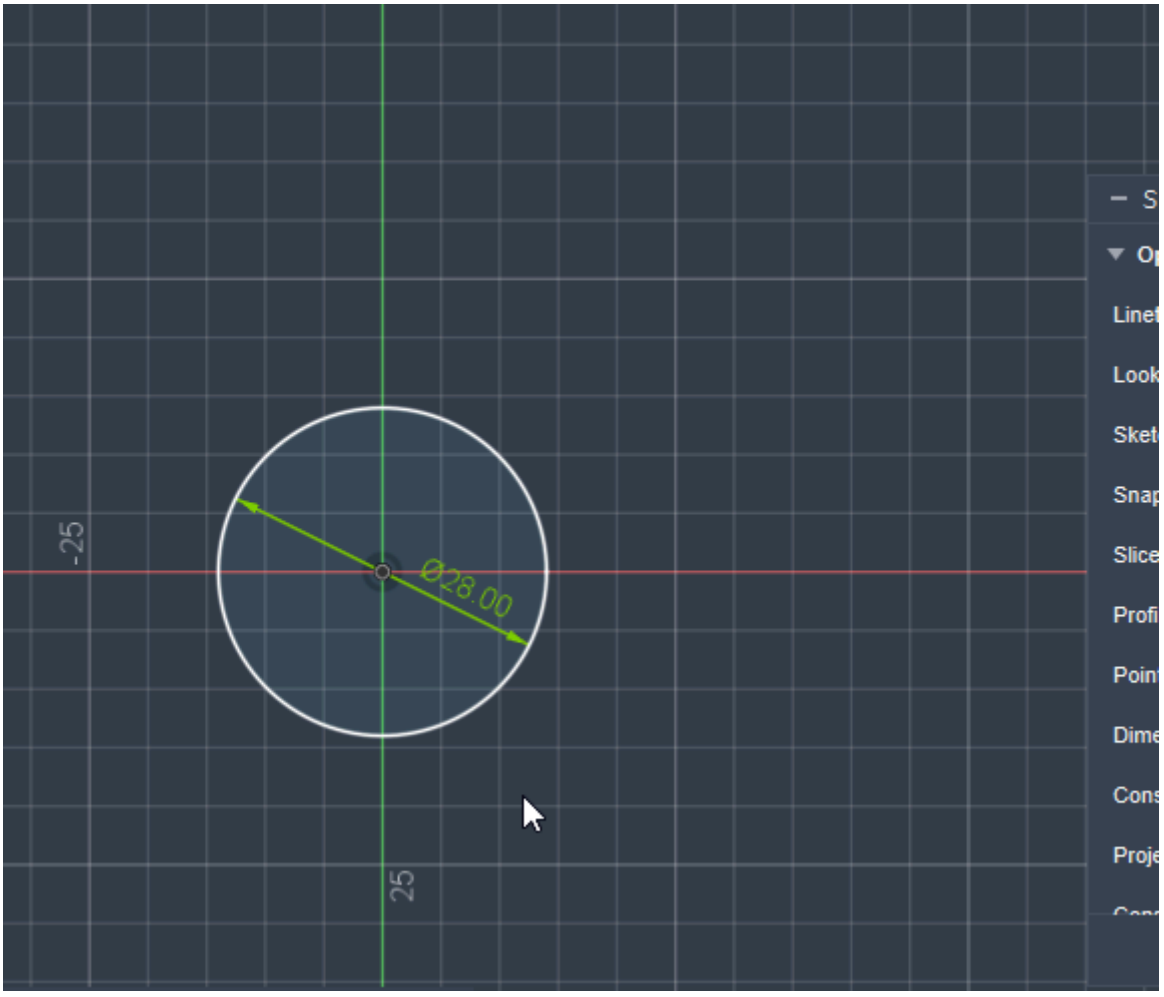
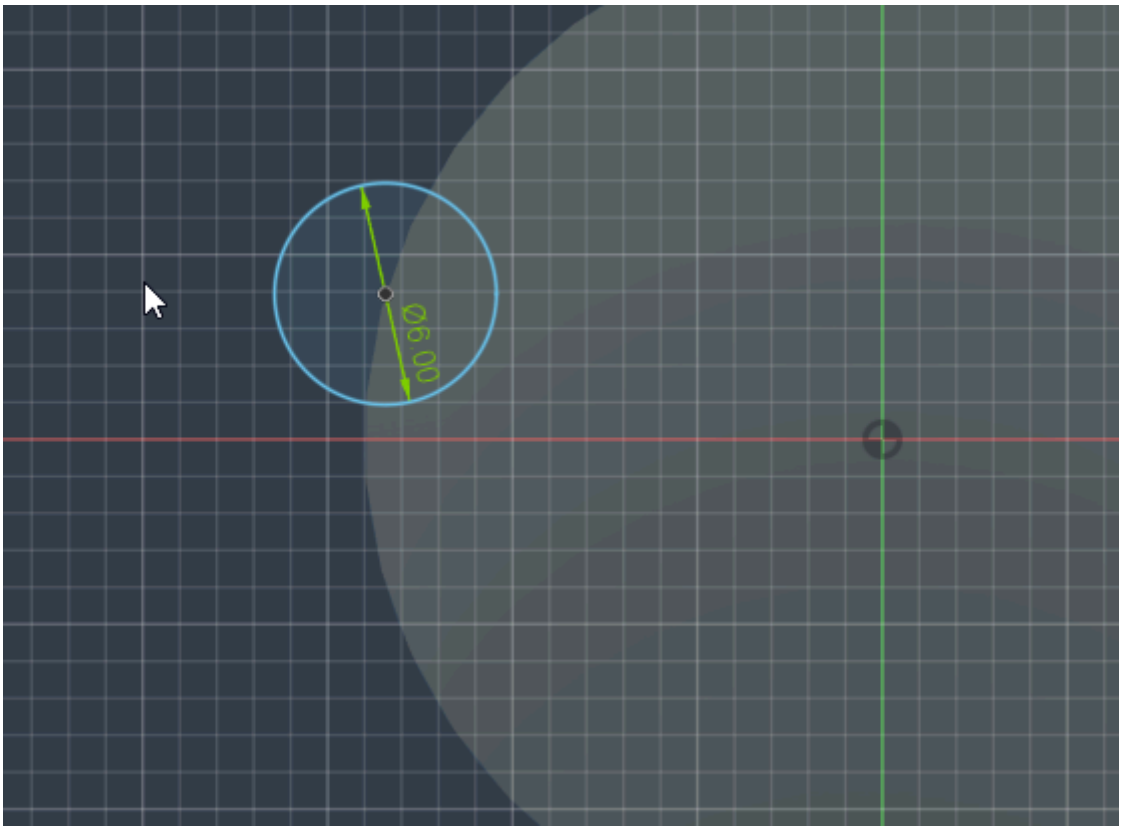
Flathead Screwdriver



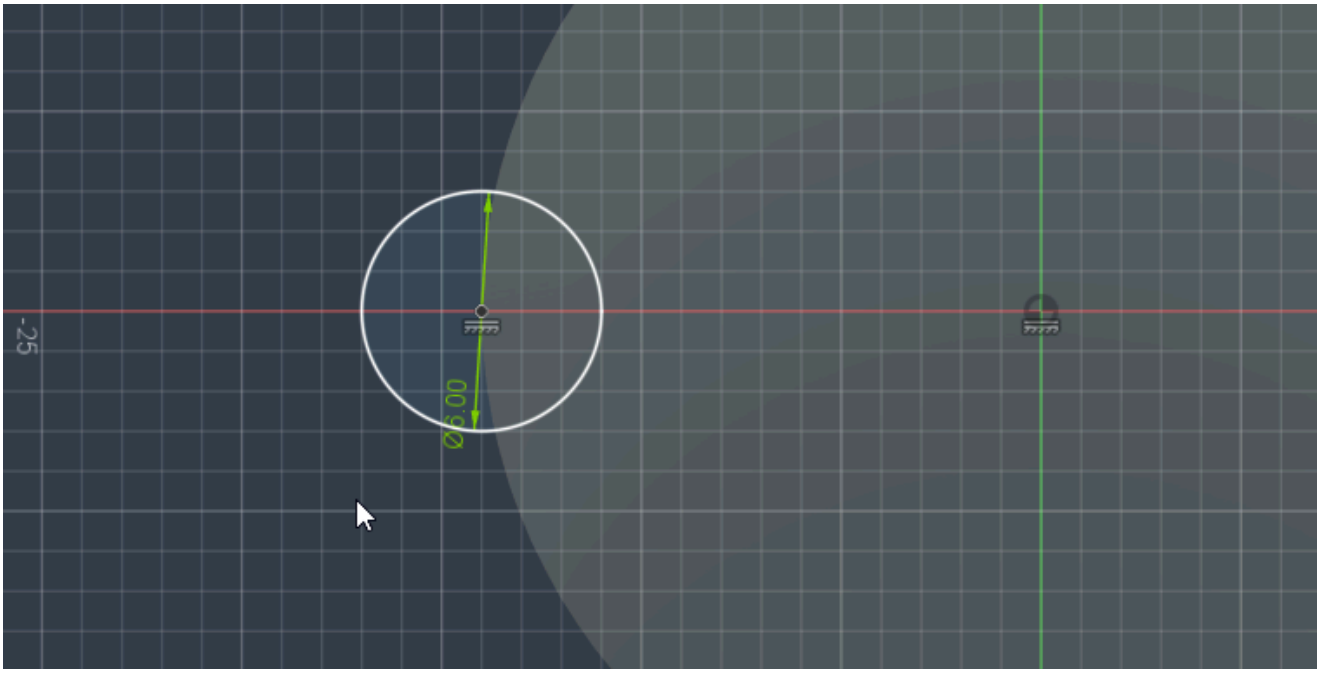
Create 3 components: Handle, Shank, Tip. Start with the Handle (XZ plane).



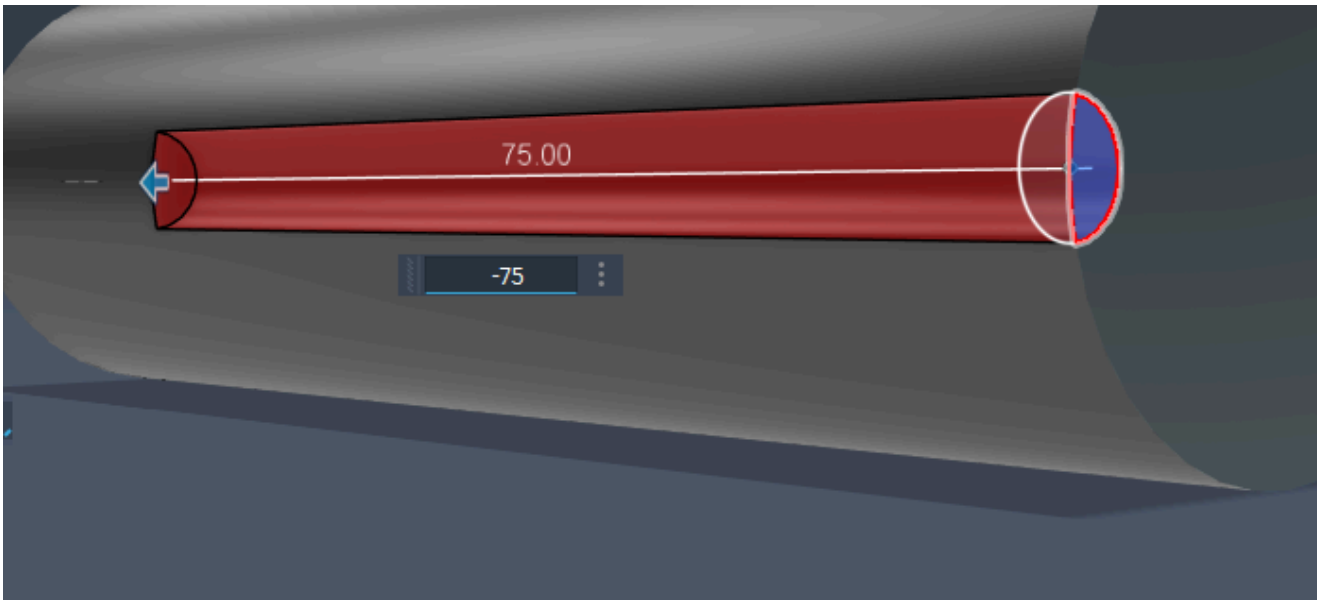
Sketch center circle: Ø28 mm → Extrude 100 mm (handle body).



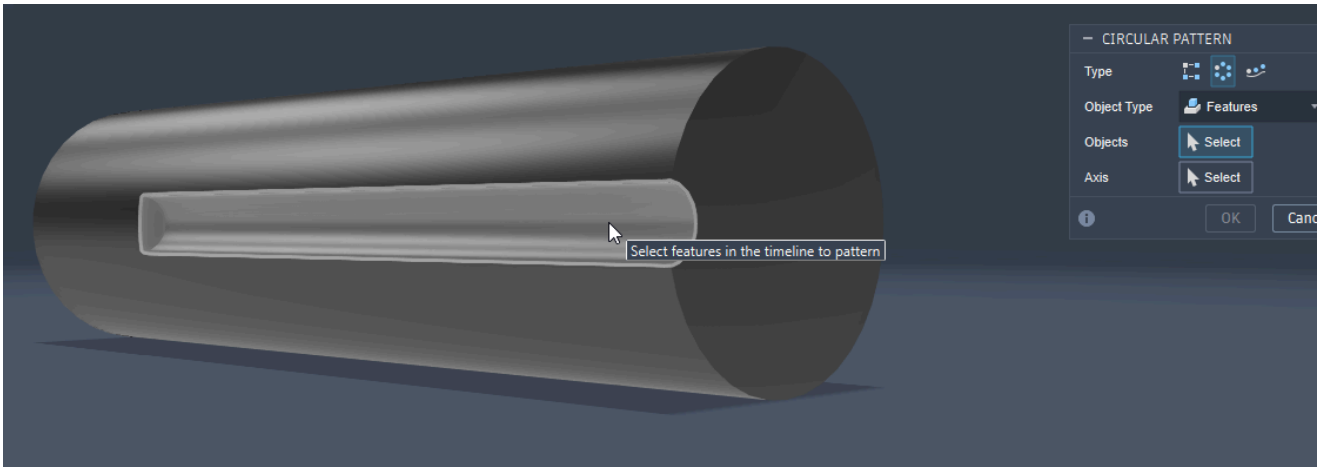
On the back face of the extruded handle, create a Ø6 mm circle (for groove/feature).



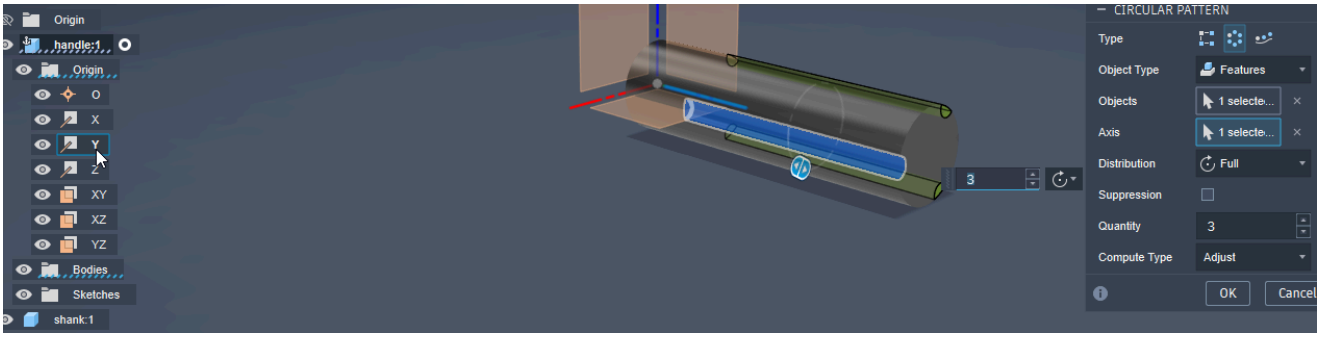
Apply a Horizontal/90° constraint to lock the position (align the small circle).



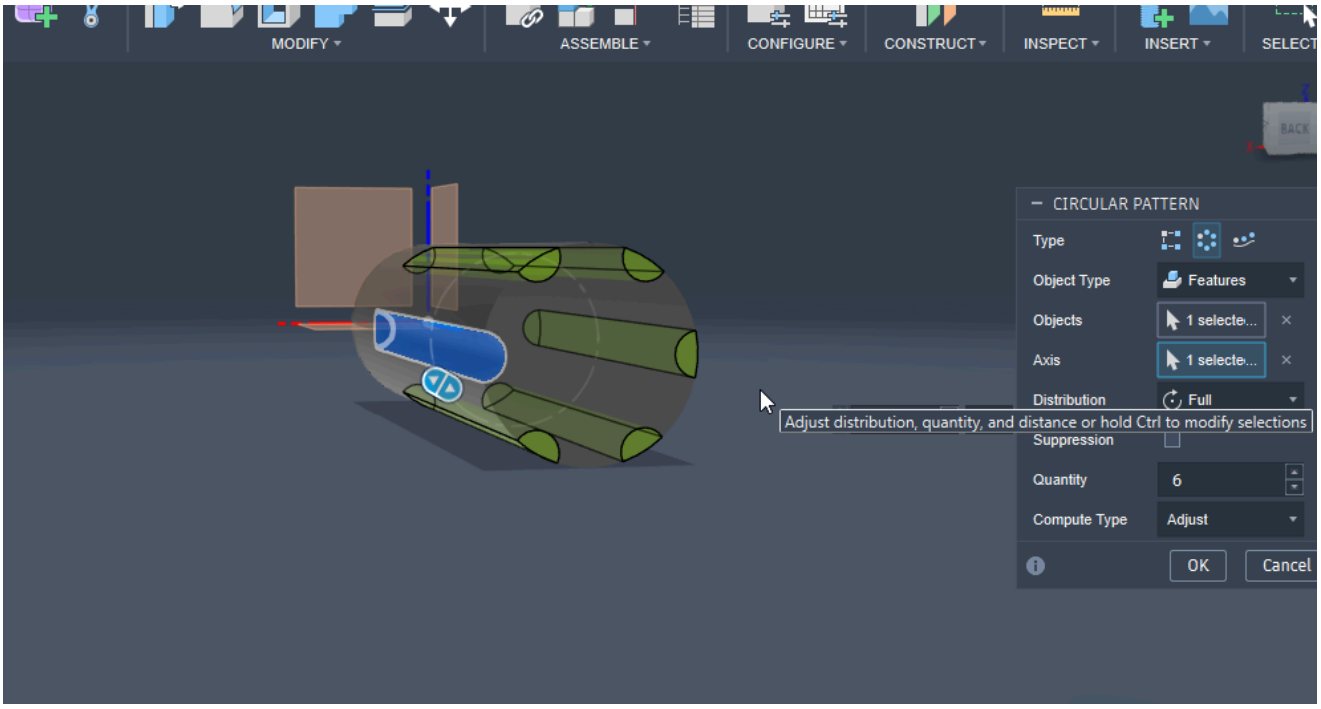
Cut-extrude the small circle into the handle: Depth = 75 mm (cut -75 mm).



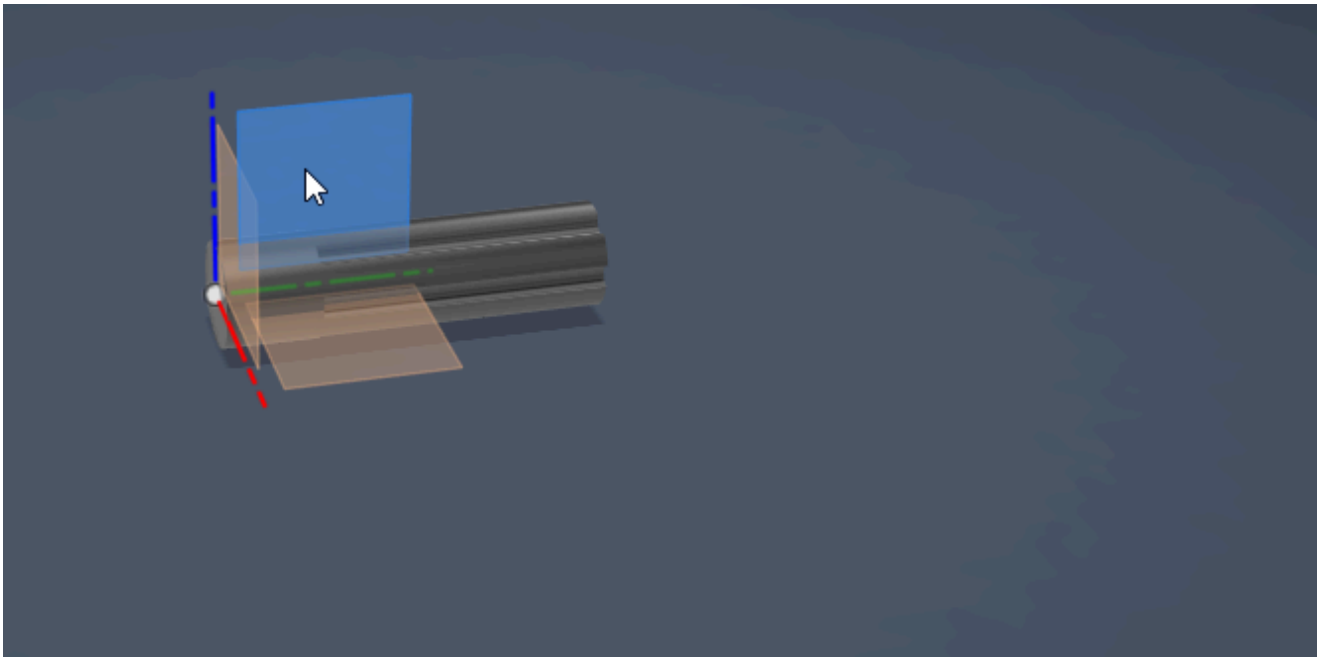
Use Circular Pattern (Features) to replicate the groove cut around handle.



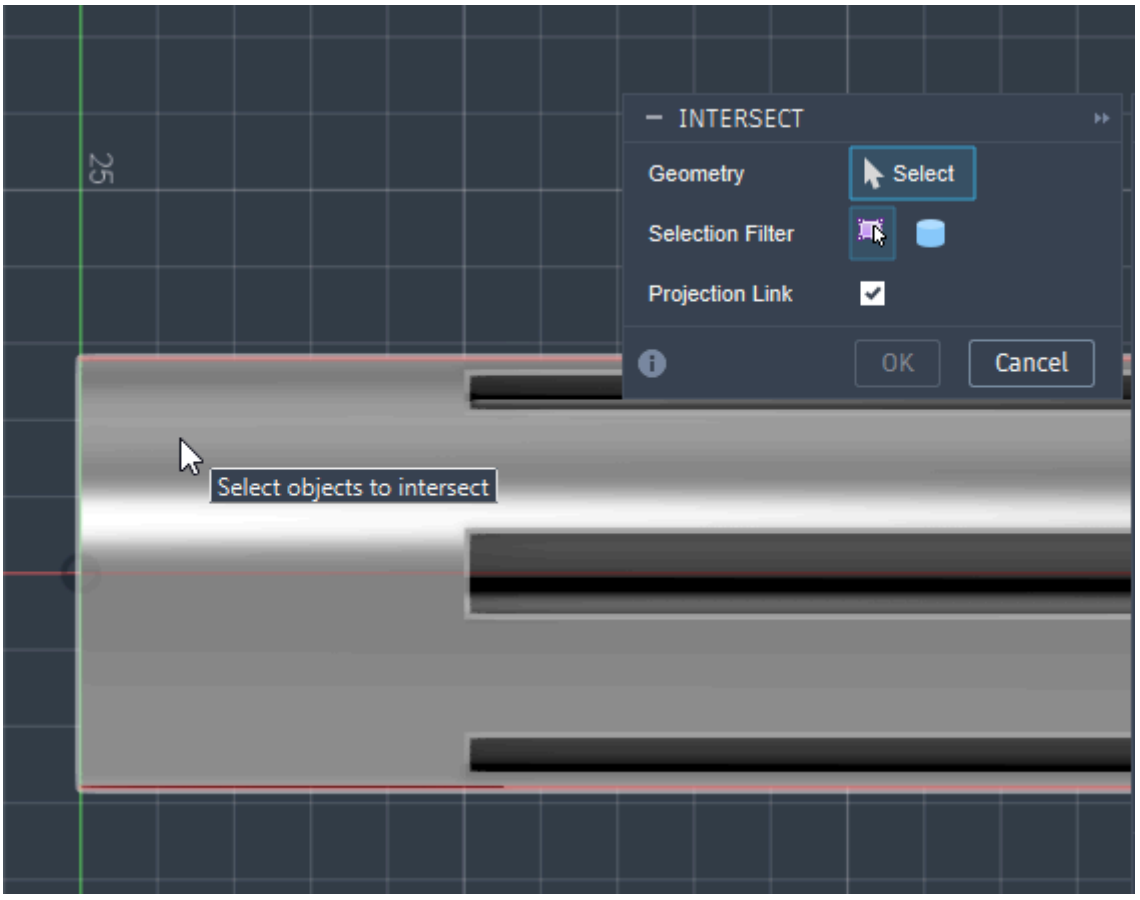
Select the Y axis as the pattern axis.



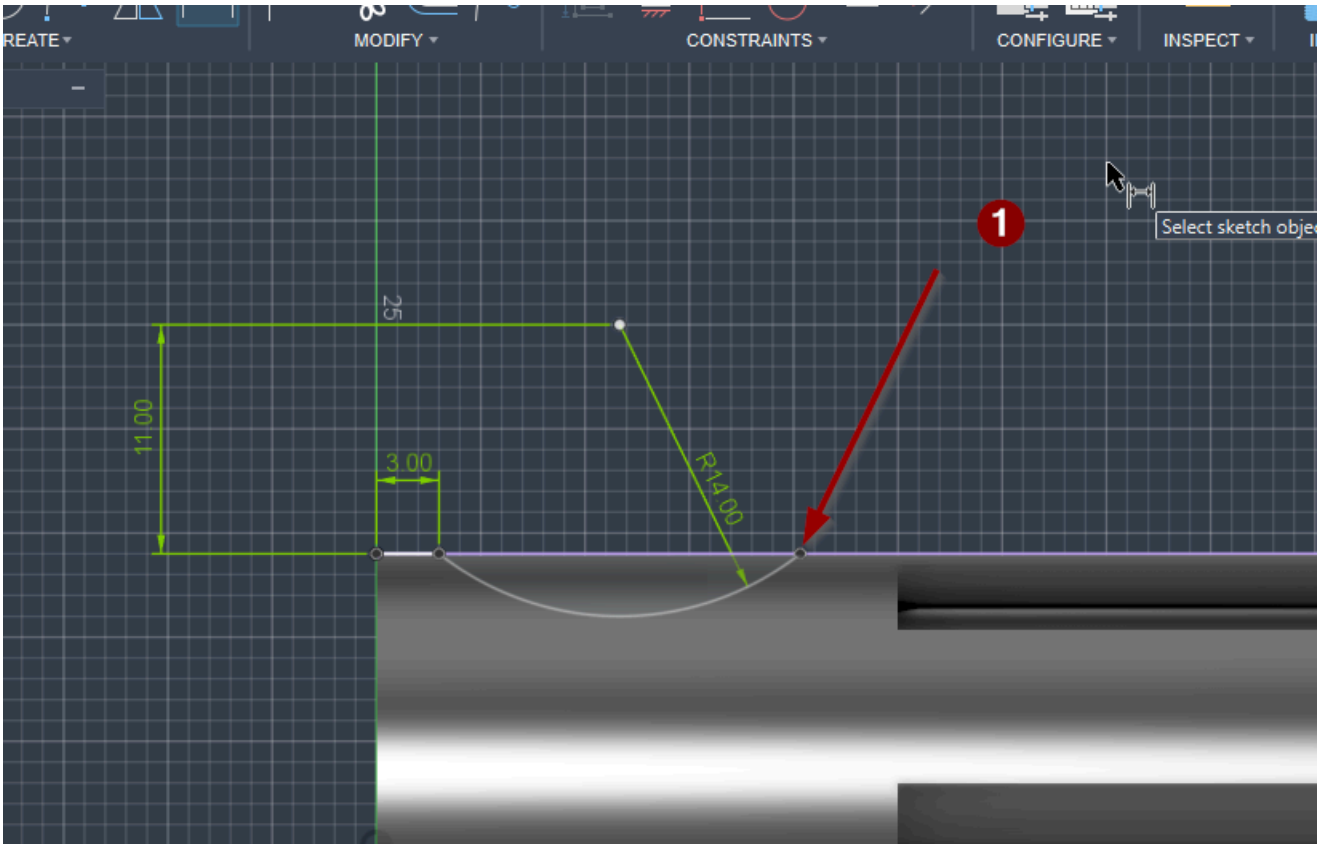
Set Quantity = 6 (six grooves around handle).



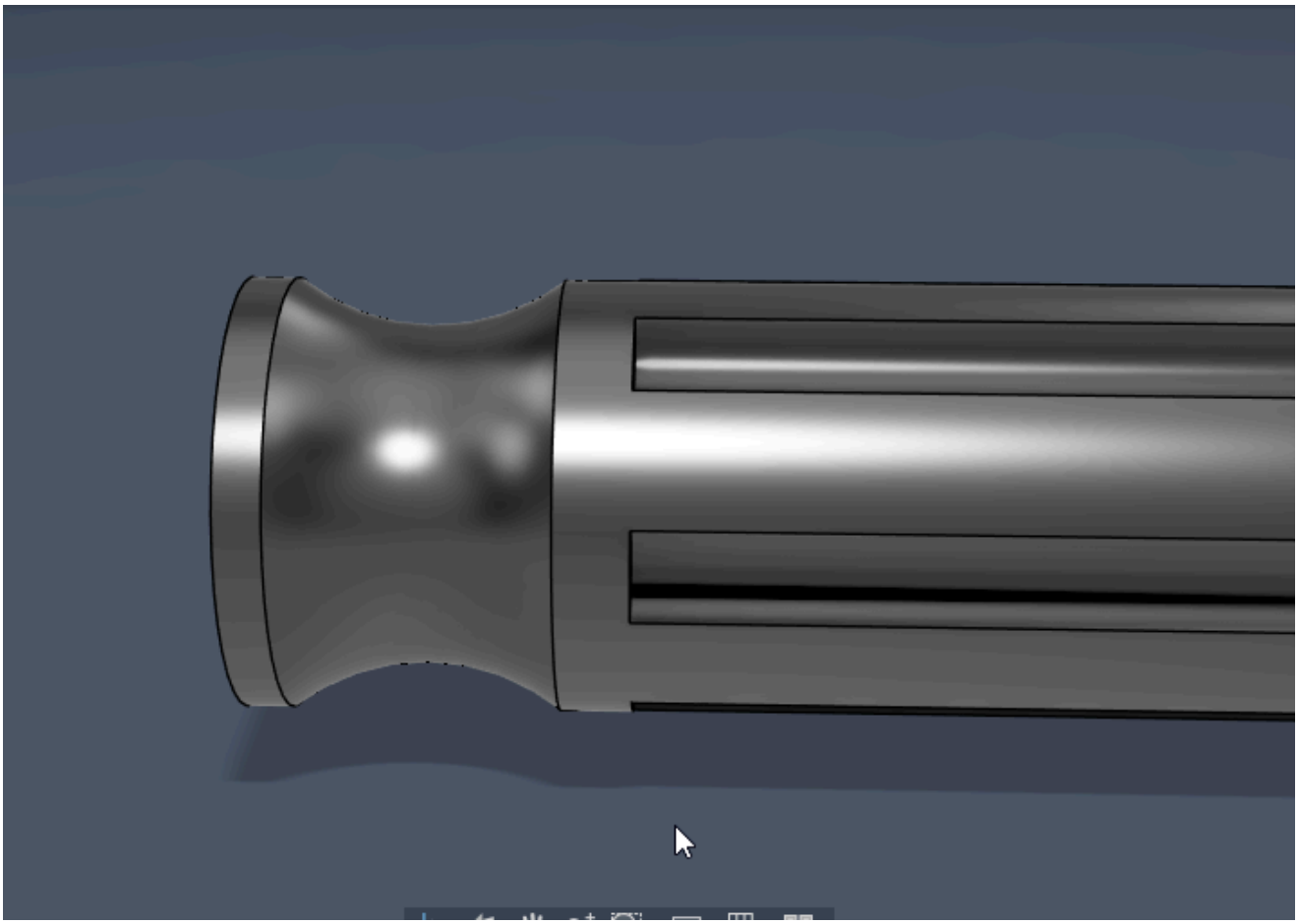
Create sketch on the YZ plane for the grip/profile features.



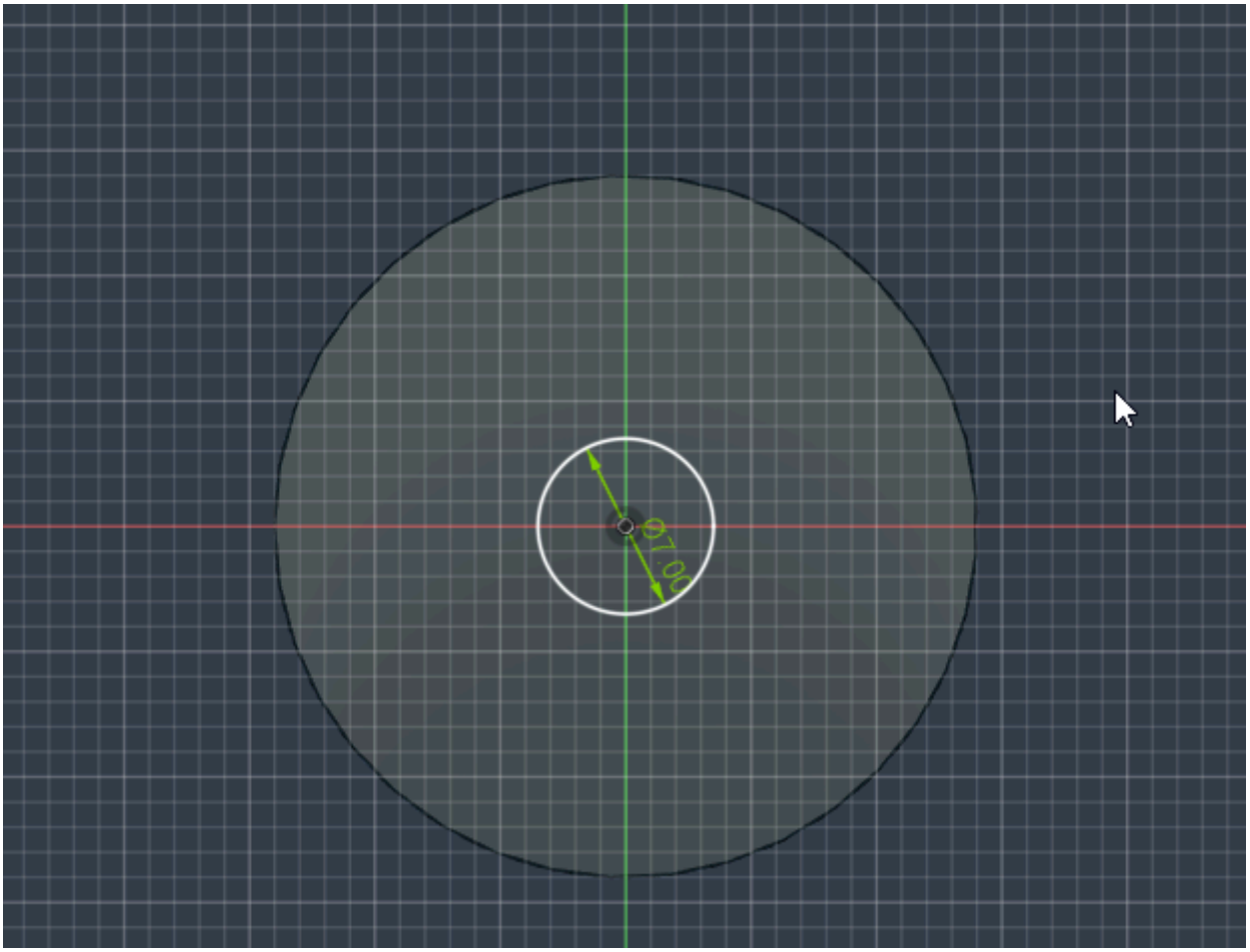
Activate Intersect (Project/Include → Intersect) to reference existing geometry into the sketch.



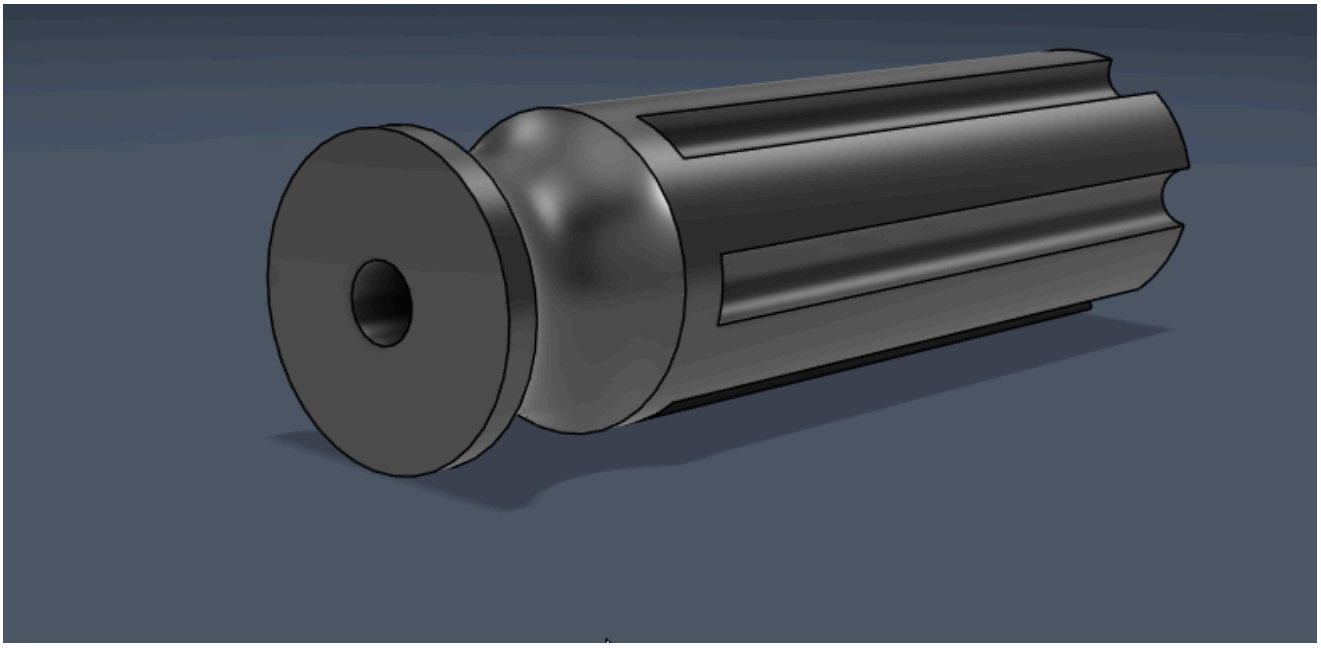
After Intersect, add dimensions and create a 3-Point Arc for the profile.



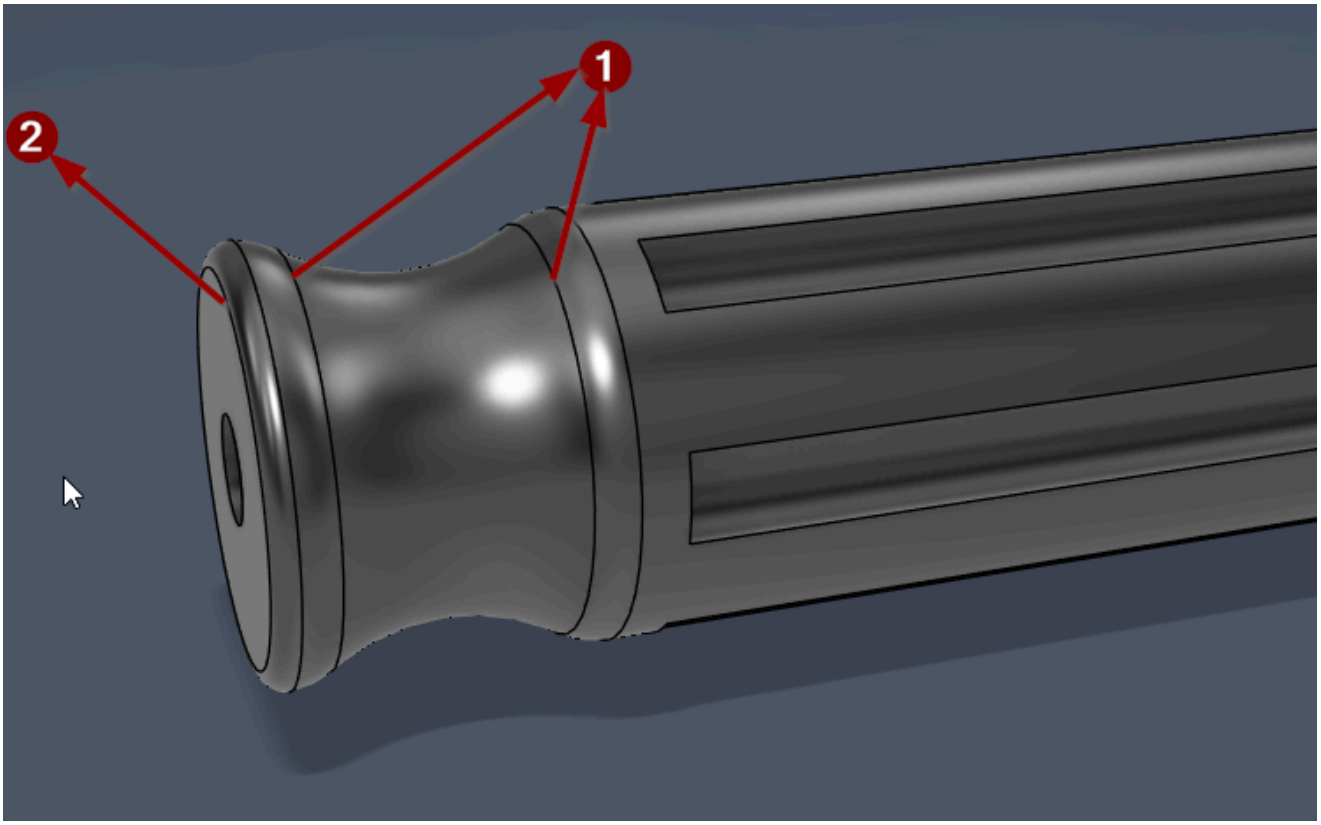
Use Revolve Cut to remove material (cut the divet/profile using revolve).



On the front face: draw Ø7 mm circle (center hole).

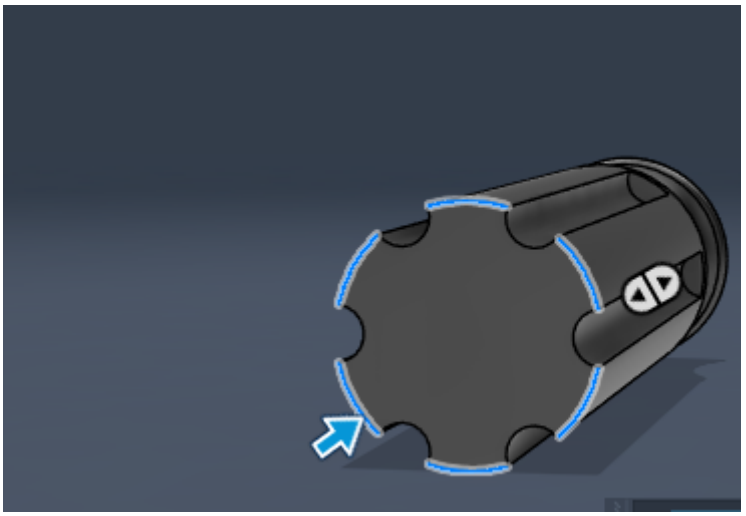


Cut-extrude the Ø7 mm hole: Depth = 50 mm (cut -50 mm).

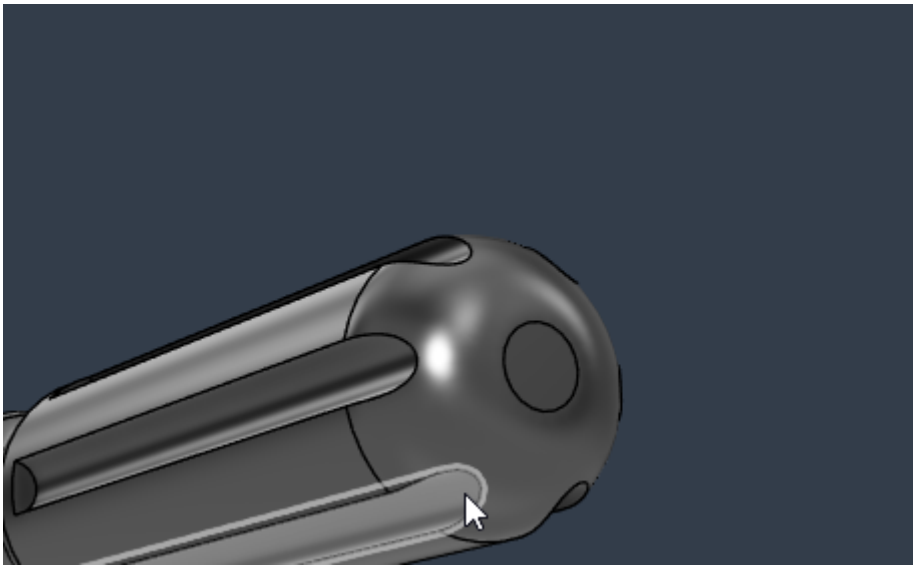


**Fillets:

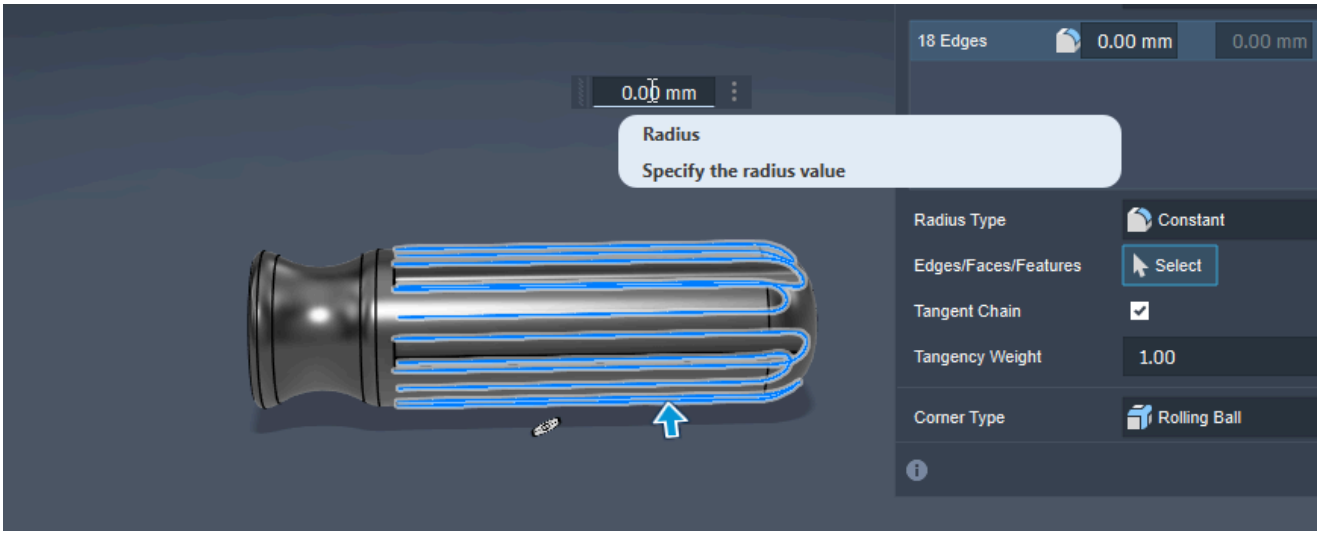
1. 5 mm fillet on divet edges
2. 1.5 mm fillet on front edge.**

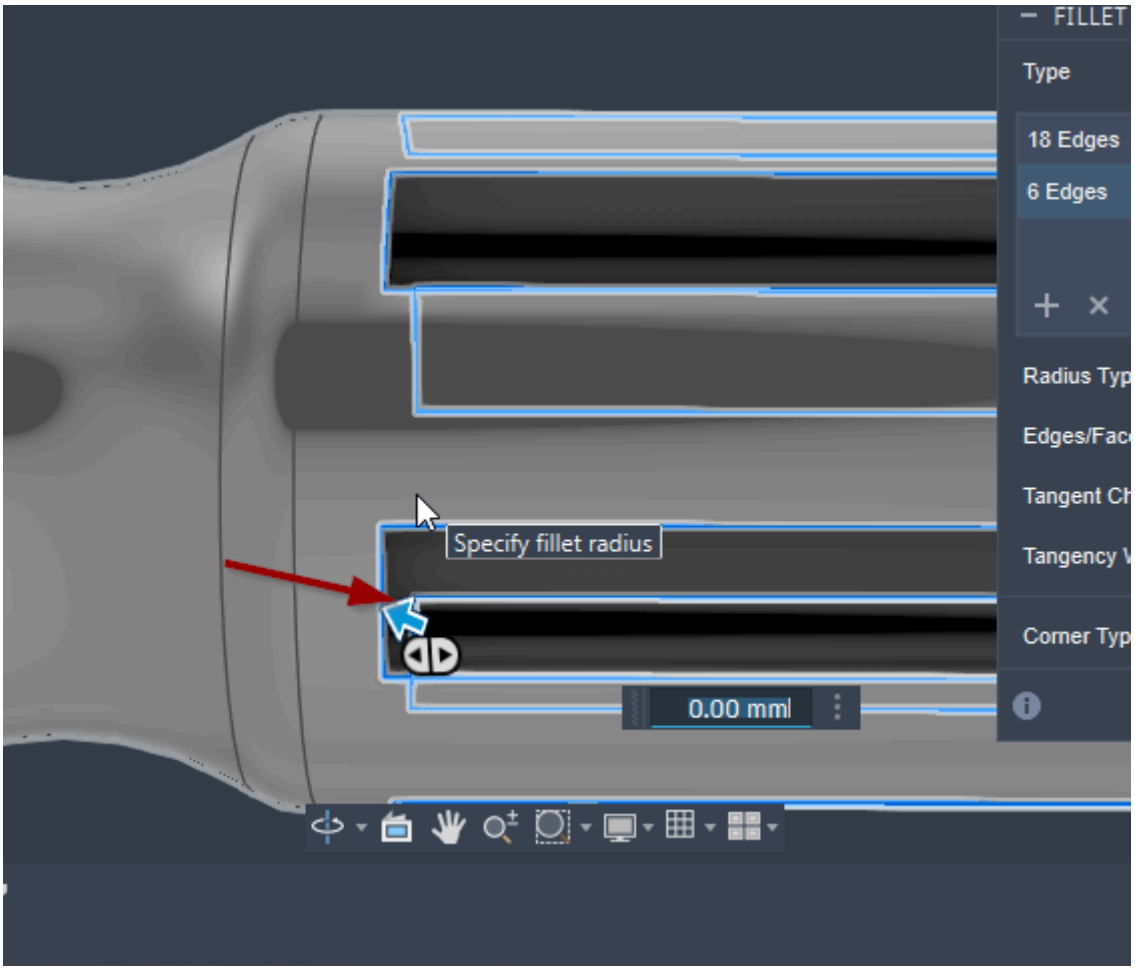


Add a larger 100 mm fillet (big roundover on the handle rear) — result shown.

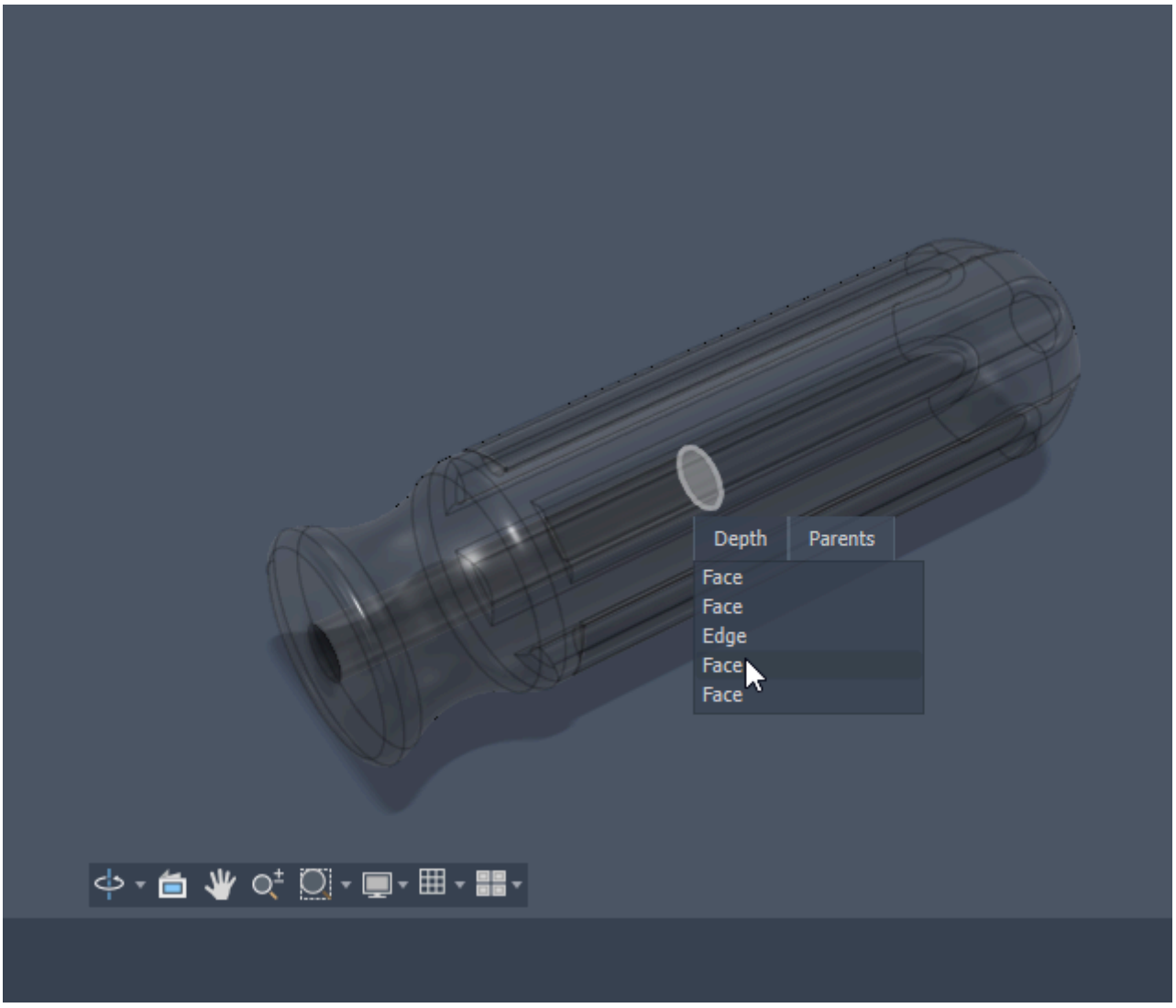


Apply 2 mm fillet to long groove edges (visual smoothing).

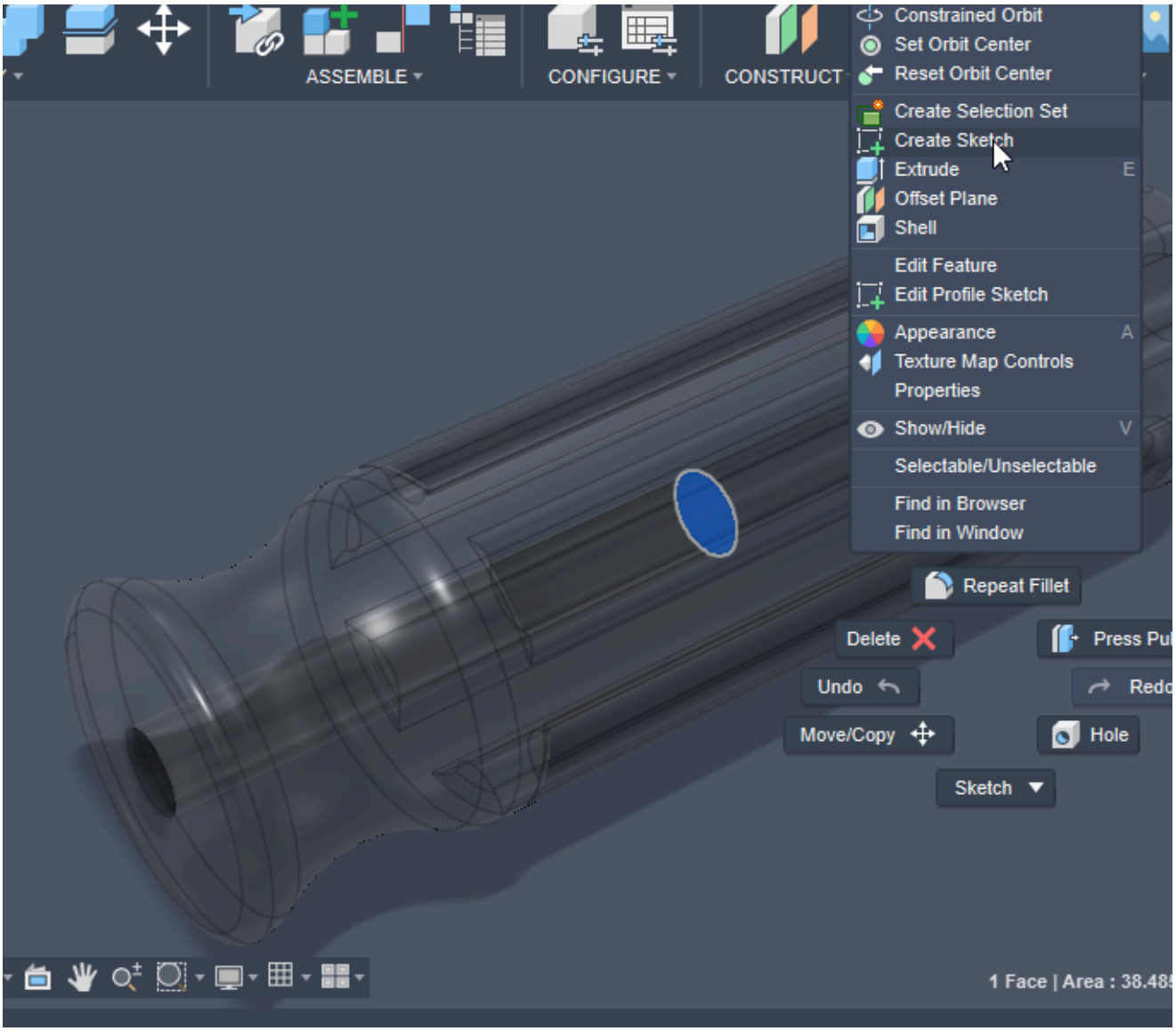




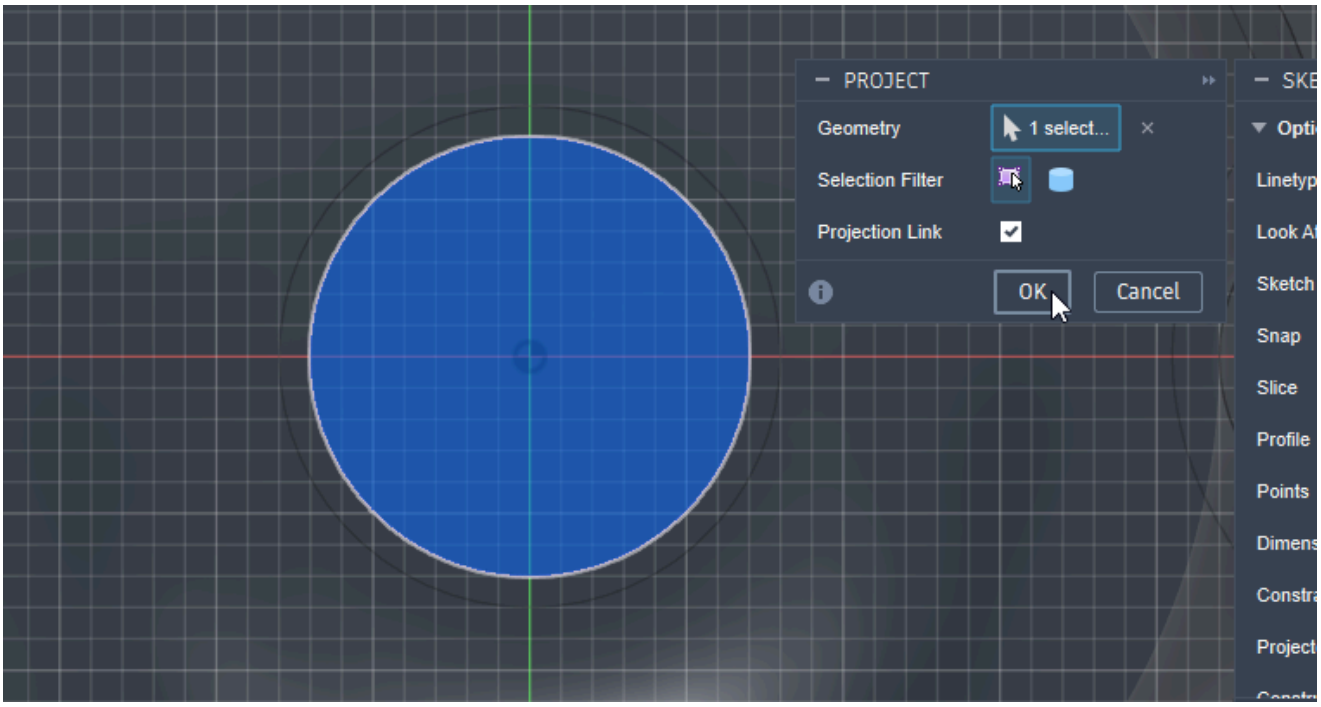
Add 1 mm fillet to small groove edges (final smoothing).



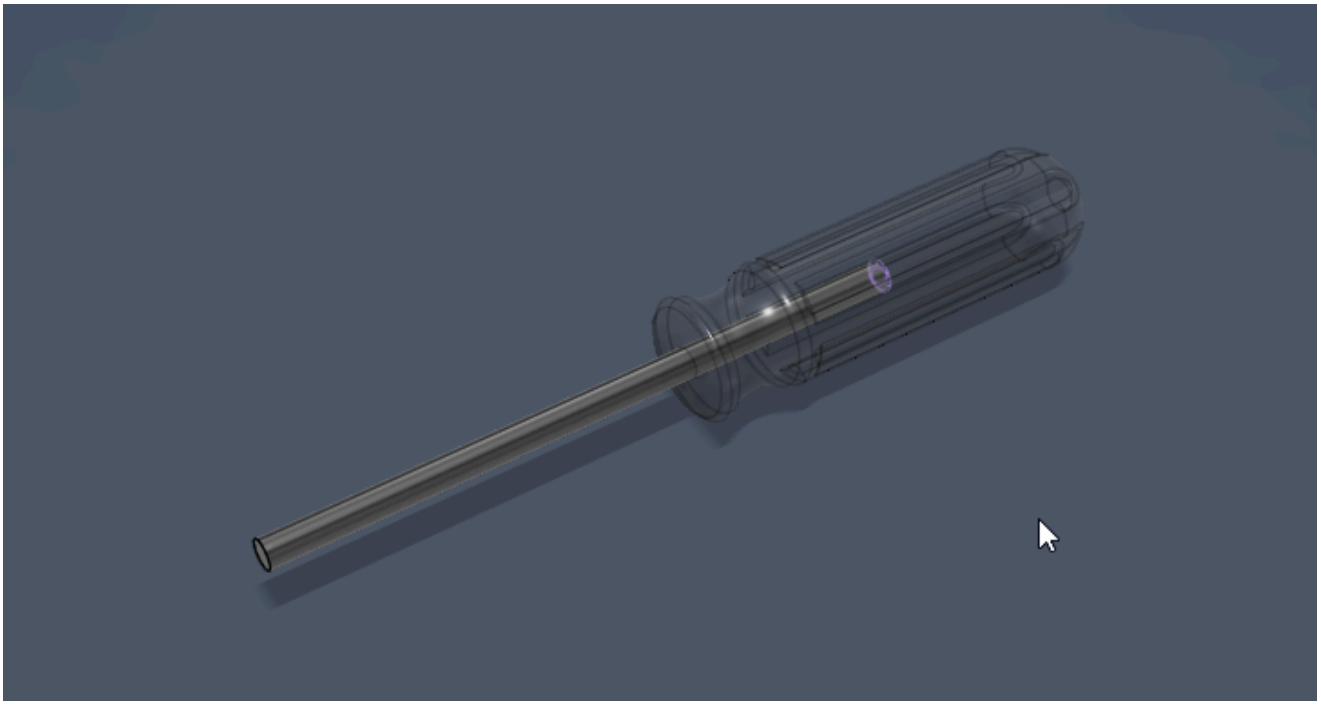
Tip: long-press (left click hold) over a face to show selection list — pick the face to create sketches from.
Right-click → Create Sketch on selected face for easy modification of the shank interface.



Use Project (P) to project that face so changes to handle diameter update the shank automatically.



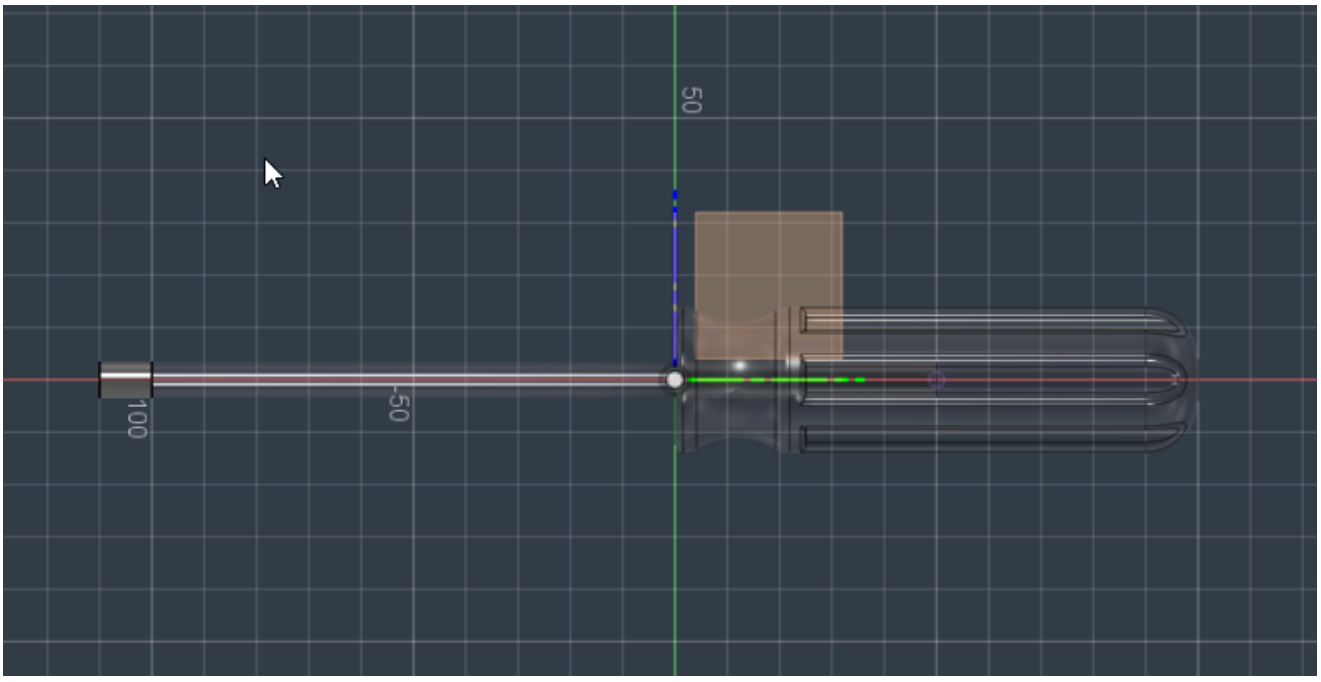
Create the Shank: Cylinder 150 mm long (extrude).



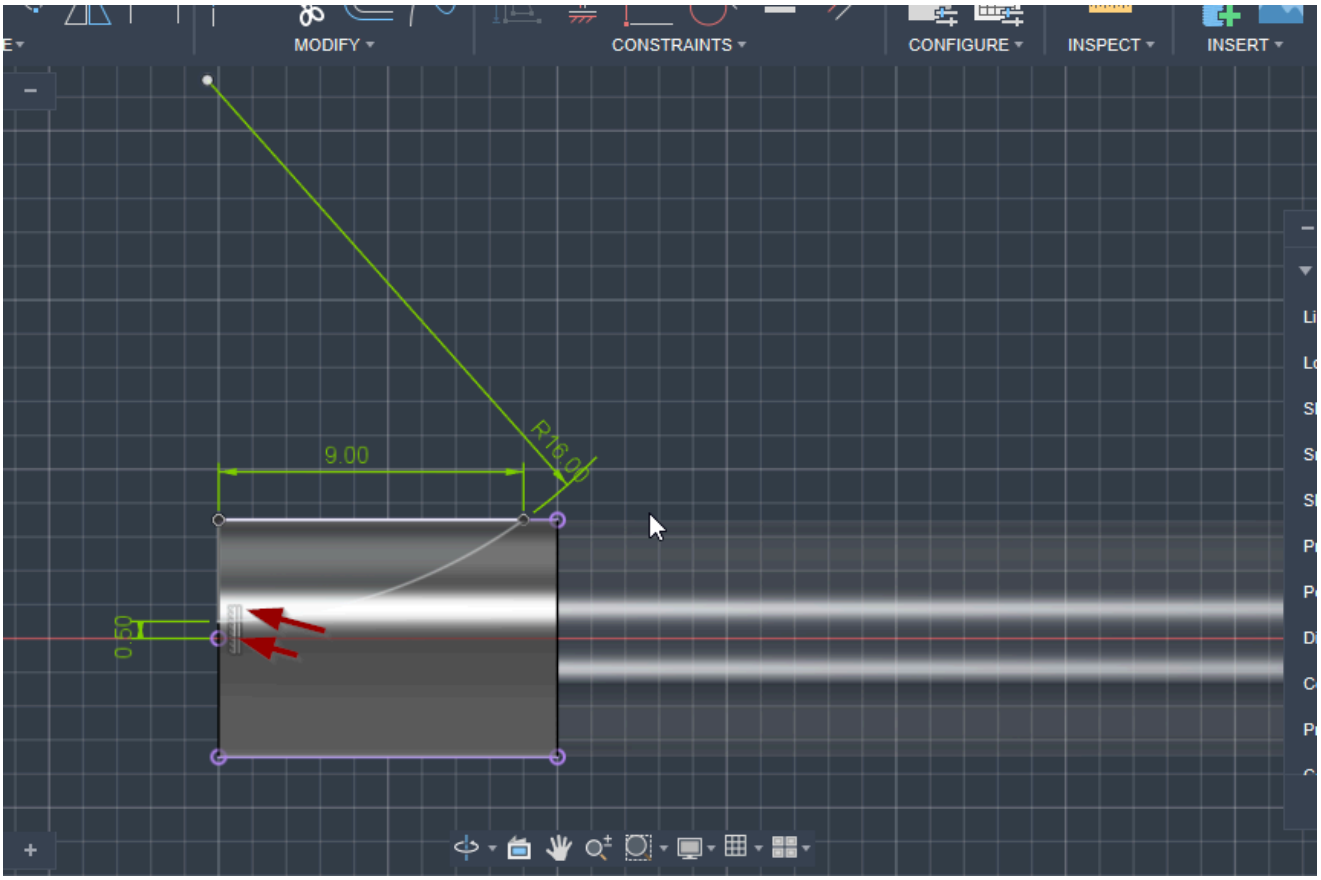
Create the Tip: extrude 10 mm from the shank end (tip body).



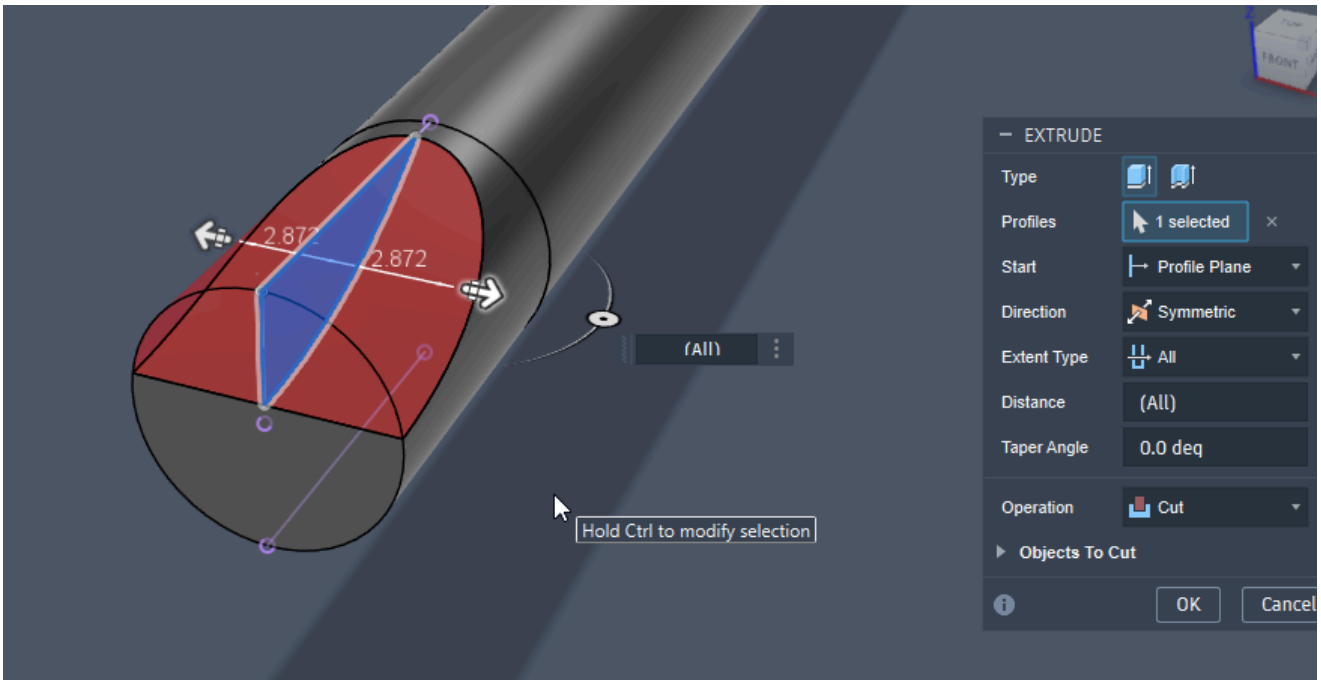
Switch to YZ plane → Start sketch for tip profile.



Use Intersect to bring edges into sketch; draw the 3-point arc profile and lock with horizontal/vertical constraints.



Cut-extrude the tip profile (use Symmetric / All as needed to remove material).



Mirror the tip cut to the other side (Solid → Mirror → Features; use XY origin plane as mirror plane).